

GENERAL PROPERTIES OF MANY PARTICLE SYSTEMS AT LOW TEMPERATURES

NISHTHA TIKALAL

CONTENTS

1	The Fermi Liquid	1
1.1	Excitations in a Fermi Liquid	1
1.2	Properties of the Excitations of Normal Fermi Liquids	2
2	Energy of the Quasi Particles	2
3	Sound	6

1. THE FERMI LIQUID

1.1. Excitations in a Fermi Liquid

Since particles will now be obeying Fermi statistics, note that spin now = 1/2. A theory of the weakly excited states of a Fermi liquid is the assumption that the excitation spectrum of the Fermi liquid is formed by the same principle as the spectrum of an ideal Fermi gas.

In momentum space the occupied states form a sphere of radius p_0 ; this is called the Fermi sphere. p_0 is called the Fermi momentum, it is determined from the condition that the number of states of $p < p_0$ should equal the number of particles:

$$p_0 = \left(3\pi^2 \underbrace{\frac{N}{V}}_{\text{particle density}} \right)^{\frac{1}{3}} \quad (1)$$

Elementary excitations differ from the original state by the presence of a particle $p_0 < p$ & a “hole” $p_0 > p$. The energy of these excitations from the surface of the Fermi sphere is given by:

$$\xi = \frac{p^2}{2m} - \frac{p_0^2}{2m} \approx v(p - p_0) \quad (2)$$

where $v = \frac{p_0}{m}$.

The energy of excitations of the “hole”-type is:

$$-\xi = \frac{p_0^2}{2m} - \frac{p^2}{2m} \approx v(p_0 - p) \quad (3)$$

Recall that we noted states of the Fermi liquid resembled states of the Fermi gas. Similar to the liquid, we can say excitations in the gas has two types:

- i) particles: $p > p_0$
- ii) holes: $p < p_0$

Its also important to note that not all the time will we be treating ideal Fermi gases & liquids as the same thing. Recall that excitations may interact w/ each other in a liquid.

A noted example is superfluid Fermi liquids. There is no superfluid Fermi gas.

1.2. Properties of the Excitations of Normal Fermi Liquids

Recall that we only discuss elementary excitation in the case where its attenuation is small compared to the energy. Attenuation is determined by:

- i) one excitation decays into several others
- ii) collision of excitations w/ each other.

To understand the properties of the energy spectrum of a Fermi liquid, we use an analogous case: The ideal Fermi gas.

Say the ground state of a system corresponds to a set of quasi-particles filling a Fermi sphere w/ momentum p_0 . Recall that in such systems, p_0 is equal to the number of particles, given by Eq. (1).

Letting the distribution function be denoted by $n(\mathbf{p})$, changes in $n(\mathbf{p})$ are subjected to the condition:

$$\int \delta n d\mathbf{p} = 0 \quad (4)$$

2. ENERGY OF THE QUASI PARTICLES

One other assumption to note regarding the interactions of quasi-particles is that it can be described as a field acting on one particle due to the others surrounding it. Thus, the energy is no longer equivalent to a sum of individual particles, but a functional of the distribution functions.

Definition 2.1. Functional: a way to turn a function into a number.

ex: $F[f]$ is a functional acting on f

$$F[f] = \int_0^1 f(x) dx$$

let $f(x) = x^2$. Then

$$F[f] = \int_0^1 x^2 dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}.$$

The energy is defined as the variational derivative of the total energy E w/ respect to the distribution function:

$$\delta E = 2V \int \varepsilon \delta n \frac{d\mathbf{p}}{(2\pi)^3} \quad (5)$$

Here, ε is the change in energy due to one extra quasi-particle w/ a momentum $\mathbf{p}$. The following note is important to avoid any restrictions: we require spatial homogeneity.

In the presence of a magnetic field (or ferromagnetic system), the distribution function is regarded as an operator acting on the spin indices as a density matrix $n_{\alpha\beta}$. The energy of the quasi particles is also an operator $\varepsilon_{\alpha\beta}$.

Equation (5) becomes

$$\delta\left(\frac{E}{V}\right) = \sum_{\alpha,\beta} \int \varepsilon_{\alpha\beta} \delta n_{\alpha\beta} \frac{d\mathbf{p}}{(2\pi)^3} \quad (6)$$

A more convenient way to write this expression is

$$\delta\left(\frac{E}{V}\right) = \text{Sp}_\sigma \int \varepsilon \delta n \frac{d\mathbf{p}}{(2\pi)^3} \quad (7)$$

Sp_σ is the trace, which is summation of diagonal components of the matrix product $\varepsilon \delta n$.

Equation (7) implies that the equilibrium distribution function is actually the Fermi distribution. It is most convenient to show this using the following expression:

$$\frac{S}{V} = -\text{Sp}_\sigma \int [n \ln n + (1-n) \ln(1-n)] \frac{d\mathbf{p}}{(2\pi)^3} \quad (8)$$

Note that S is just the entropy. Recall the condition of entropy that it is a maximum when the # of particles & energy is conserved. This occurs at

$$\delta N = 0, \quad \delta E = 0.$$

Under this condition, the distribution function is

$$n(\varepsilon) = n_F(\varepsilon) = \frac{1}{e^{(\varepsilon-\mu)/T} + 1} \quad (9)$$

ε is given by

$$\begin{aligned} \varepsilon(\mathbf{p}, \boldsymbol{\sigma}) &= \underbrace{\varepsilon^{(0)}(\mathbf{p}, \boldsymbol{\sigma})}_{\substack{\text{equilibrium} \\ \text{energy @ } T=0}} + \delta\varepsilon(\mathbf{p}, \boldsymbol{\sigma}) \\ &= \varepsilon^{(0)}(\mathbf{p}, \boldsymbol{\sigma}) + \text{Sp}_{\sigma'} \int f(\mathbf{p}, \boldsymbol{\sigma}; \mathbf{p}', \boldsymbol{\sigma}') \delta n(\mathbf{p}', \boldsymbol{\sigma}') \frac{d\mathbf{p}'}{(2\pi)^3}. \end{aligned} \quad (10)$$

Here,

$$\delta n = n - n_F(T=0).$$

f is an operator depending on spin and momentum. Note that

$$\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z),$$

which we know to be the Pauli matrices. They are related to spin by

$$\mathbf{s} = \frac{1}{2} \boldsymbol{\sigma}. \quad (11)$$

Here, $\delta n = n - n_F(T=0)$. f is an operator depending on spin / momentum. Note that $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$, which we know to be Pauli matrices! They are related to spin by:

$$\mathbf{s} = \frac{1}{2} \boldsymbol{\sigma} \quad (12)$$

A more detailed version of Eq. (10) explains the meaning of $f(\mathbf{p}, \boldsymbol{\sigma}; \mathbf{p}', \boldsymbol{\sigma}')$

$$\varepsilon_{\alpha\beta}(\mathbf{p}) = \varepsilon_{\alpha\beta}^{(0)}(\mathbf{p}) + \underbrace{\int f_{\alpha\beta,\gamma\delta}(\mathbf{p}, \mathbf{p}') \delta n_{\delta\gamma}(\mathbf{p}') \frac{d\mathbf{p}'}{(2\pi)^3}}_{\text{second variational derivative w/ respect to } \delta n \text{ of the energy per unit volume.}} \quad (13)$$

This is telling us that f is symmetric under the permutation $\mathbf{p}, \boldsymbol{\sigma} \rightleftharpoons \mathbf{p}', \boldsymbol{\sigma}'$. The dependence of f on the spin variables can be expressed as:

$$f(\mathbf{p}, \boldsymbol{\sigma}; \mathbf{p}', \boldsymbol{\sigma}') = \varphi(\mathbf{p}, \mathbf{p}') + \sigma_i \sigma'_k \zeta_{ik}(\mathbf{p}, \mathbf{p}') \quad (14)$$

In the absence of magnetic fields, ε is independent of spin. Therefore,

$$\varepsilon^{(0)}(\mathbf{p})$$

in Eq. (10) is only $\mathbf{p}$ dependent, and can be expanded in a series $(p - p_0)$:

$$\xi(p) = \varepsilon^{(0)}(p) - \mu(0) \approx v(p - p_0) \quad (15)$$

where $\mu(0)$ is the chemical potential at $T = 0$, and v is a constant. Specifically, v is the velocity of excitations at the Fermi surface, and can be expressed as:

$$v = \frac{p_0}{m^*} \quad (16)$$

There is a connection between f and the effective mass m^* . To understand, note that the momentum of a unit volume of the liquid is the same as the flow of mass. The momentum of a unit volume of the Fermi liquid is the same as the momentum of quasi-particles:

$$\mathbf{P}(\mathbf{p}) = 2 \int \mathbf{p} n \frac{d\mathbf{p}}{(2\pi)^3}$$

Because of this the current of the fluid particles coincides with the current of the quasi-particles:

$$\mathbf{J}(\mathbf{v}) = 2 \int \mathbf{v} n \frac{d\mathbf{p}}{(2\pi)^3}$$

The density of the mass current is obtained from the above expression by simply multiplying it with the mass m of an atom of the liquid. Note that

$$\mathbf{v} = \nabla_{\mathbf{p}} \varepsilon,$$

allowing us to write the condition of equality between momentum and the corresponding flow of mass:

$$\int \mathbf{p} n \frac{d\mathbf{p}}{(2\pi)^3} = m \int (\nabla_{\mathbf{p}} \varepsilon) n \frac{d\mathbf{p}}{(2\pi)^3} \quad (17)$$

We can manipulate Eq. (15) w/ respect to n . Recall that changes in energy is associated w/ δn by Eq. (10):

$$\delta \varepsilon = \frac{1}{2} \text{Sp}_{\sigma} \text{Sp}_{\sigma'} \int f(\mathbf{p}, \boldsymbol{\sigma}; \mathbf{p}', \boldsymbol{\sigma}') \delta n' \frac{d\mathbf{p}'}{(2\pi)^3} \quad (18)$$

This takes prevalence when there is no magnetic field. n and ε become spin-independent. By Eq. (10) and Eq. (16), Eq. (15) becomes:

$$\int \mathbf{p} \delta n \frac{d\mathbf{p}}{(2\pi)^3} = \int (\nabla_{\mathbf{p}} \varepsilon) \delta n \frac{d\mathbf{p}}{(2\pi)^3} + \frac{1}{2} \text{Sp}_{\sigma} \text{Sp}_{\sigma'} \int n \delta n' \nabla_{\mathbf{p}} f(\mathbf{p}, \sigma; \mathbf{p}', \sigma') \frac{d\mathbf{p} d\mathbf{p}'}{(2\pi)^6}$$

To evaluate, our focus is applied towards the second term (or integrand). Recall that the expression for integration by parts is:

$$\int u dv = uv - \int v du$$

Let

$$u = n = n(p); \quad du = \frac{\partial n}{\partial p} dp.$$

Let

$$\frac{\partial}{\partial p} f(\mathbf{p}, \sigma; \mathbf{p}', \sigma') dp = dv;$$

therefore,

$$v = f(\mathbf{p}, \sigma; \mathbf{p}', \sigma').$$

By the condition that

$$\mathbf{p}, \sigma \rightleftharpoons \mathbf{p}', \sigma',$$

which makes f symmetric, we can interchange the regular variables with the dummy variables. Therefore, the full expression becomes:

$$\int \frac{\mathbf{p}}{m} \delta n \frac{d\mathbf{p}}{(2\pi)^3} = \int (\nabla_{\mathbf{p}} \varepsilon) \delta n \frac{d\mathbf{p}}{(2\pi)^3} - \frac{1}{2} \text{Sp}_{\sigma} \text{Sp}_{\sigma'} \int \delta n f(\mathbf{p}, \sigma; \mathbf{p}', \sigma') \nabla_{\mathbf{p}'} n' \frac{d\mathbf{p} d\mathbf{p}'}{(2\pi)^6}$$

Since δn is arbitrary, the expression becomes:

$$\frac{\mathbf{p}}{m} = \nabla_{\mathbf{p}} \varepsilon - \frac{1}{2} \text{Sp}_{\sigma} \text{Sp}_{\sigma'} \int f(\mathbf{p}, \sigma; \mathbf{p}', \sigma') \nabla_{\mathbf{p}'} n' \frac{d\mathbf{p}'}{(2\pi)^3}$$

At $T = 0$, the energy ε near the Fermi surface takes the form of Eq. (13), while

$$\nabla_{\mathbf{p}'} n' \approx -\frac{\mathbf{p}'}{p_0} \delta(p' - p_0)$$

Since f only depends on the angle χ between $\mathbf{p}$ and $\mathbf{p}'$ (due to the isotropy of the liquid),

$$\frac{1}{m^*} = \frac{1}{m} - \frac{p_0}{2(2\pi)^3} \text{Sp}_{\sigma} \text{Sp}_{\sigma'} \int f(\chi) \cos \chi d\Omega \quad (19)$$

$f(\chi)$ is the value of f for $|\mathbf{p}| = |\mathbf{p}'| = p_0$. $d\Omega$ is the element of solid angle.

The heat capacity of a Fermi liquid can be expressed in terms of m^* by the formula of a Fermi gas. From Eq. (5), the heat capacity per unit volume is:

$$C_v = \left(\frac{\partial(E/V)}{\partial T} \right)_{N,V} = 2 \int \varepsilon \left(\frac{\partial n}{\partial T} \right)_N \frac{d\mathbf{p}}{(2\pi)^3} \quad (20)$$

If we replace ε with $\varepsilon^{(0)}$ in Eq. (18), we face a relative error. To the approximation linear in T , the heat capacity is given by the same formula:

$$C_v = \frac{1}{3} m^* p_0 T \quad (21)$$

3. SOUND

Sound propagation differs between Fermi & Bose liquids. At “not too low” temperatures sound propagation will follow hydrodynamical laws. Here, attenuation is $\propto$ to the time τ between collisions of excitations. As the temperature decreases like the square of the “spread” of the Fermi distribution, the collision time increases by T^{-2} .

When τ is the same order as ω^{-1} , no sound propagation is experienced. If we lower the temperature again, sound can once again propagate, but with a different velocity.

From this, the point stands that the relation between ω and τ determines the nature of sound. There are two kinds of characterization to note:

- i) $\omega\tau \ll 1$; low frequency sound
- ii) $\omega\tau \gg 1$; high frequency sound

When condition (i) is satisfied, the velocity of sound u can be determined by compressibility.

u is dependent on f . We show this in terms of $\frac{\partial\mu}{\partial N}$, with μ being the chemical potential and N the number of particles.

Recall that the chemical potential is dependent on N/V . Also recall the Helmholtz free-energy differential at constant temperature:

$$F(T, V, N) : \quad dF = -P dV + \mu dN$$

From this equation,

$$P = -\left(\frac{\partial F}{\partial V}\right)_{T,N} ; \quad \mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}$$

Applying Clairaut’s theorem of mixed partial derivatives, we obtain:

$$\begin{aligned} \frac{\partial^2 F}{\partial N \partial V} &= \frac{\partial^2 F}{\partial V \partial N} \\ \frac{\partial}{\partial N} \left(\frac{\partial F}{\partial V} \right) &= \frac{\partial}{\partial V} \left(\frac{\partial F}{\partial N} \right) \\ -\left(\frac{\partial P}{\partial N} \right)_V &= \left(\frac{\partial \mu}{\partial V} \right)_N \end{aligned}$$

This is our Maxwell relation.

To determine the density dependence, observe how both μ and P change with N and V .

$$\left(\frac{\partial \mu}{\partial N} \right)_V = \frac{\partial \mu}{\partial n} \left(\frac{\partial n}{\partial N} \right)_V = \frac{\partial \mu}{\partial n} \left(\frac{1}{V} \right)$$

$$\begin{aligned}
\left(\frac{\partial\mu}{\partial V}\right)_N &= \frac{\partial\mu}{\partial n} \left(\frac{\partial n}{\partial V}\right)_N \\
&= \frac{\partial\mu}{\partial n} \left[\frac{\partial}{\partial V}(NV^{-1})\right] \\
&= \frac{\partial\mu}{\partial n} \left(-\frac{N}{V^2}\right)
\end{aligned}$$

Relating our two results,

$$\frac{\partial\mu}{\partial n} \left(\frac{1}{V}\right) = \frac{\partial\mu}{\partial n} \left(-\frac{N^2}{V}\right)$$

which gives

$$\left(\frac{\partial\mu}{\partial V}\right)_N = -\frac{N}{V} \left(\frac{\partial\mu}{\partial N}\right)_V$$

Substituting this back into the Maxwell relation,

$$-\left(\frac{\partial P}{\partial N}\right)_V = -\frac{N}{V} \left(\frac{\partial\mu}{\partial N}\right)_V$$

and dividing out the negative,

$$\left(\frac{\partial P}{\partial N}\right)_V = \frac{N}{V} \left(\frac{\partial\mu}{\partial N}\right)_V$$

We now repeat this process to see how P varies with N and V . Rewriting the PDEs by the chain rule,

$$\begin{aligned}
\frac{\partial P}{\partial N} &= \frac{\partial P}{\partial n} \left(\frac{1}{V}\right) \\
\frac{\partial P}{\partial V} &= \frac{\partial P}{\partial n} \left(-\frac{N}{V^2}\right) \implies -\frac{N}{V} \cdot \frac{1}{V} \\
\frac{\partial P}{\partial N} &= -\frac{V}{N} \left(\frac{\partial P}{\partial V}\right)
\end{aligned}$$

Relating this to our previously derived relation,

$$\begin{aligned}
-\frac{V}{N} \left(\frac{\partial P}{\partial V}\right) &= \frac{N}{V} \left(\frac{\partial\mu}{\partial N}\right) \\
-\frac{V^2}{N^2} \left(\frac{\partial P}{\partial V}\right) &= \left(\frac{\partial\mu}{\partial N}\right) \\
\frac{\partial\mu}{\partial N} &= -\frac{V^2}{N^2} \left(\frac{\partial P}{\partial V}\right) = -\frac{V^2}{N^2} \left(-\frac{N^2}{V} \frac{\partial P}{\partial(N/V)}\right) = \frac{1}{N} \frac{\partial P}{\partial(N/V)}
\end{aligned} \tag{22}$$

This implies the relation:

$$u^2 = \frac{\partial P}{\partial \rho} = \frac{\partial P}{\partial(mN/V)} = \frac{1}{m} N \frac{\partial\mu}{\partial N} \tag{23}$$

In order to calculate $\frac{\partial \mu}{\partial N}$, let

$$\mu \approx \varepsilon(p_0).$$

Thus changes in μ are due both to changes in p_0 and changes in the shape of the function $\varepsilon(p)$:

$$\delta \mu = \frac{1}{2} \text{Sp}_\sigma \text{Sp}_{\sigma'} \int f \delta n' \frac{d\mathbf{p}'}{(2\pi)^3} + \frac{\partial \varepsilon^{(0)}(p_0)}{\partial p_0} \delta p_0 \quad (24)$$

By Eq. (1), changes in δN and δp_0 are related by:

$$\delta N = \frac{1}{\pi^2} p_0^2 \delta p_0 V$$

Since changes in $\delta n'$ near the Fermi surface take precedence in Eq. (22), integration is carried out over the magnitude of momentum:

$$\int f' \delta n' \frac{d\mathbf{p}'}{(2\pi)^3} = \frac{\delta N}{8\pi V} \int f d\Omega$$

By the result above,

$$\frac{\partial \mu}{\partial N}$$

becomes:

$$\frac{\partial \mu}{\partial N} = \frac{1}{16\pi V} \text{Sp}_\sigma \text{Sp}_{\sigma'} \int f d\Omega + \frac{\pi^2}{p_0 m^* V} \quad (25)$$

By Eq. (17) for the effective mass and Eq. (1), Eq. (21) becomes:

$$u^2 = \frac{p_0^2}{3m^2} + \frac{1}{6} \left(\frac{p_0}{2\pi} \right)^3 \text{Sp}_\sigma \text{Sp}_{\sigma'} \int f(\chi) (1 - \cos \chi) d\Omega \quad (26)$$

This allows us the velocity of sound in the regime of $\omega\tau \ll 1$. Without interactions,

$$u^2 = \frac{p_0^2}{3m^2}$$

In the frequency range of $\omega\tau \gg 1$, the transport equation is applied:

$$\frac{\partial n}{\partial t} + \nabla_{\mathbf{r}} n \cdot \nabla_{\mathbf{p}} \varepsilon - \nabla_{\mathbf{p}} n \cdot \nabla_{\mathbf{r}} \varepsilon = I(n) \quad (27)$$

where $I(n)$ is the collision integral. For small deviation from equilibrium, we note the distribution function as:

$$n = n_F + \delta n,$$

where n_F is the equilibrium distribution and δn is a small segment, represented as a periodic function in time:

$$\delta n \sim e^{i(\mathbf{k} \cdot \mathbf{r} - \omega T)}$$

The collision integral $I(n)$ is of order

$$\frac{\delta n}{\tau},$$

so this can be neglected compared to the term

$$\frac{\partial n}{\partial t}.$$

One important note regarding Eq. (25) is that

$$\nabla_{\mathbf{r}}\varepsilon \neq 0.$$

By Eq. (10),

$$\nabla_{\mathbf{r}}\varepsilon = \text{Sp}_{\sigma'} \int f \nabla_{\mathbf{r}} \delta n' \frac{d\mathbf{p}'}{(2\pi)^3}$$

From this, we obtain:

$$(\mathbf{v} \cdot \mathbf{k} - \omega) \delta n - (\mathbf{v} \cdot \mathbf{k}) \frac{\partial n_F}{\partial \varepsilon} \text{Sp}_{\sigma'} \int f \delta n' \frac{d\mathbf{p}'}{(2\pi)^3} = 0 \quad (28)$$

Eq. (26) is obtained through Eq. (25). Recall that:

$$n = n_F + \delta n; \quad \delta n \sim e^{i(\mathbf{k} \cdot \mathbf{r} - \omega T)}$$

and

$$\begin{cases} \mathbf{v} \cdot \nabla_{\mathbf{r}} \delta n = i(\mathbf{v} \cdot \mathbf{k}) \delta n, \\ \nabla_{\mathbf{p}} n(\mathbf{p}) = \mathbf{v} \frac{\partial n_F}{\partial \varepsilon}. \end{cases}$$

These values allow us to attain Eq. (26). From this equation, we see that

$$\delta n \propto \frac{\partial n_F}{\partial \varepsilon} \approx -\delta(\varepsilon - \mu).$$

If we let

$$\delta n = \frac{\partial n_F}{\partial \varepsilon} \nu,$$

then Eq. (26) becomes:

$$(\mathbf{v} \cdot \mathbf{k} - \omega) \nu + (\mathbf{v} \cdot \mathbf{k}) \frac{1}{2} \text{Sp}_{\sigma'} \int f \nu' \frac{d\Omega'}{4\pi} = 0 \quad (29)$$

Note that:

$$F(\chi) = f(\chi) \frac{p_0 m^*}{\pi^2} \quad (30)$$

Choosing $\mathbf{k}$ as the polar axis and denoting the velocity of the wave propagation by

$$\tilde{u} = \frac{\omega}{k},$$

then Eq. (27) becomes:

$$(s - \cos \theta) \nu(\theta, \varphi, \boldsymbol{\sigma}) = \cos \theta \frac{1}{2} \text{Sp}_{\sigma'} \int F(\chi) \nu(\theta', \varphi', \boldsymbol{\sigma}') \frac{d\Omega'}{4\pi} \quad (31)$$

From Eq. (29),

$$s = \frac{\tilde{u}}{v}.$$

So, in the regime of $\omega\tau \gg 1$, sound propagating through a Fermi liquid causes the Fermi surface to no longer remain spherical, but instead change in a complex way. Changes to the Fermi surface are determined by ν .

To study this, we first look at the spin-independent solution to Eq. (29). Here, from all of $F(\chi)$, the only remaining term is $\Phi(\chi)$, which is connected with φ from Eq. (12).

Consider:

$$\Phi = \Phi_0 = \text{constant}.$$

From Eq. (29),

$$\nu = \frac{\cos \theta}{s - \cos \theta} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega T)} \quad (32)$$

Substitution of Eq. (30) into Eq. (29) and letting $F = \Phi_0$ yields an expression for s :

$$\begin{aligned} (s - \cos \theta) \nu(\theta, \varphi, \boldsymbol{\sigma}) &= \cos \theta \frac{1}{2} \text{Sp}_{\sigma'} \int F(\chi) \nu(\theta', \varphi', \boldsymbol{\sigma}') \frac{d\Omega'}{4\pi} \\ (s - \cos \theta) \frac{\cos \theta}{s - \cos \theta} \underbrace{e^{i(\mathbf{k} \cdot \mathbf{r} - \omega T)}}_{\text{let this be } \alpha}(\theta, \varphi, \boldsymbol{\sigma}) &= \cos \theta \cdot \frac{1}{2} \text{Sp}_{\sigma'} \int \frac{\Phi_0 \cos \theta}{s - \cos \theta} (\theta', \varphi', \boldsymbol{\sigma}') \frac{d\Omega'}{4\pi} \\ \alpha \cos \theta &= \cos \theta \Phi_0 \left[\frac{1}{2} \text{Sp}_{\sigma'} \int \nu(\theta', \varphi', \boldsymbol{\sigma}') \frac{d\Omega'}{4\pi} \right] \end{aligned}$$

Recall that ν is independent of $\boldsymbol{\sigma}$, so:

$$\frac{1}{2} \text{Sp}_{\sigma'}(1) = \frac{1}{2}(1 + 1) = 1.$$

Therefore:

$$\begin{aligned} \alpha \cos \theta &= \cos \theta \Phi_0 \int \frac{\alpha \cos \theta'}{s - \cos \theta'} \frac{d\Omega'}{4\pi} \\ 1 &= \Phi_0 \int \frac{\cos \theta'}{s - \cos \theta'} \frac{d\Omega'}{4\pi} \end{aligned}$$

Recall that $d\Omega'$ is the solid angle element:

$$d\Omega' = \sin \theta' d\theta' d\varphi'.$$

Therefore:

$$1 = \Phi_0 \int \frac{\cos \theta' \sin \theta'}{s - \cos \theta'} d\theta' \int_0^{2\pi} d\varphi' \frac{1}{4\pi}$$

$$1 = \frac{\Phi_0}{2} \int \frac{\cos \theta' \sin \theta'}{s - \cos \theta'} d\theta'$$

To evaluate this integrand, let

$$u = \cos \theta'.$$

Thus,

$$du = -\sin \theta' d\theta'.$$

Substitution into the integrand yields:

$$1 = -\frac{\Phi_0}{2} \int_{-1}^1 \frac{u}{s - u} du$$

Re-expressing the integrand:

$$\frac{u}{s - u} = \frac{u - s + s}{s - u} = -1 + \frac{s}{s - u}$$

Therefore:

$$1 = -\frac{\Phi_0}{2} \int_{-1}^1 \left(-1 + \frac{s}{s - u} \right) du$$

$$1 = -\frac{\Phi_0}{2} [-u + s \ln |s - u|]_{-1}^1$$

At $u = 1$:

$$-1 - s \ln(s - 1)$$

At $u = -1$:

$$1 - s \ln(s + 1)$$

Thus:

$$1 = -\frac{\Phi_0}{2} [-2 + s \ln(s + 1) - s \ln(s - 1)]$$

$$1 = -\frac{\Phi_0}{2} \left[-2 + s \ln \left(\frac{s + 1}{s - 1} \right) \right]$$

Hence:

$$\frac{s}{2} \ln \left(\frac{s + 1}{s - 1} \right) - 1 = \frac{1}{\Phi_0} \quad (33)$$

If s is real, then s must be larger than unity:

$$\tilde{u} > v$$

as shown by Eq. (32). Since the left-hand side is always positive ($s > 0$), a condition for zero sound is that Φ_0 is positive as well.

From Eq. (31),

$$\frac{s}{2} \ln \left(\frac{s+1}{s-1} \right) - 1 = \frac{1}{\Phi_0}$$

with the limiting behavior:

$$s \rightarrow \sqrt{\frac{\Phi_0}{3}} \quad \text{as } \Phi_0 \rightarrow \infty;$$

and

$$s \rightarrow 1 \quad \text{as } \Phi_0 \rightarrow 0.$$

This tells us that the velocity of zero sound can exceed the velocity of regular sound by $(\sqrt{3})^{-1}$. In the case that instead of Φ_0 , we have $\Phi(\chi)$, Eq. (29) becomes extremely complex.

In the case where f depends on spins of particles,

$$F(\chi) = \Phi(\chi) + \zeta(\chi)(\boldsymbol{\sigma} \cdot \boldsymbol{\sigma}') \quad (34)$$

Eq. (29) is satisfied by a function of:

$$\nu = \boldsymbol{\nu} \cdot \boldsymbol{\sigma} \quad (35)$$

but $\boldsymbol{\nu}$ is some unknown vector.

For this function, we obtain:

$$(s - \cos \theta) \boldsymbol{\nu} = \cos \theta \int \zeta \boldsymbol{\nu}' \frac{d\Omega'}{4\pi} \quad (36)$$

REFERENCES

- [1] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed. Cambridge, UK: Cambridge University Press, 2018.
- [2] A. A. Abrikosov, L. P. Gorkov, and I. E. Dzyaloshinski, *Methods of Quantum Field Theory in Statistical Physics*. New York, NY: Dover Publications, 1975.
- [3] T. Lancaster and S. J. Blundell, *Quantum Field Theory for the Gifted Amateur*. Oxford, UK: Oxford University Press, 2014.
- [4] S. J. Blundell and K. M. Blundell, *Concepts in Thermal Physics*, 2nd ed. Oxford, UK: Oxford University Press, 2009.